Sols

1. The purpose of this problem is to see how to construct a derivative function one point at a time by looking at a graph.
   Background review: estimating derivatives, one point at a time:

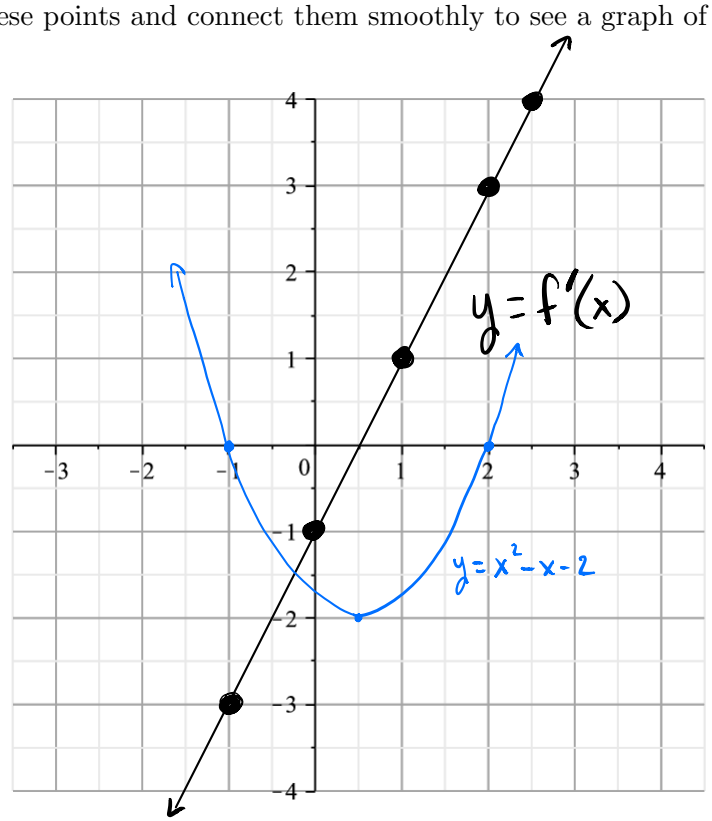
   - The derivative of a function at a point represents the slope (or rate of change) of a function at that point.
   - If you have a graph, you can estimate the derivative one point at a time by drawing the tangent line at that point, then calculating the slope of that tangent line (remember, slope is rise over run).

   (a) Go to the website http://www.shodor.org/interactivate/activities/Derivate/

   (b) Enter the function $y = x^2 - x - 2$. Use the tool to calculate the slope of the graph at each of the points x = -1, 0, 1, 2 and 2.5. Enter the values of the slope in the following table:

x	-1	0	1	2	2.5
slope of f at x	-3	-1	1	3	4

   (c) Now plot these points and connect them smoothly to see a graph of $f'(x)$



   (d) What do you think the formula for this graph is? $y = f'(x) = 2x - 1$

2. In this problem, you'll calculate the derivative of the same function as the previous problem, but this time you'll do it analytically (with formulas)

   (a) Calculate the derivative of $f(x) = x^2 - x - 2$.

   $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} =$

$$= \lim_{h \to 0} \frac{(x+h)^2 - (x+h) - 2 - (x^2 - x - 2)}{h}$$

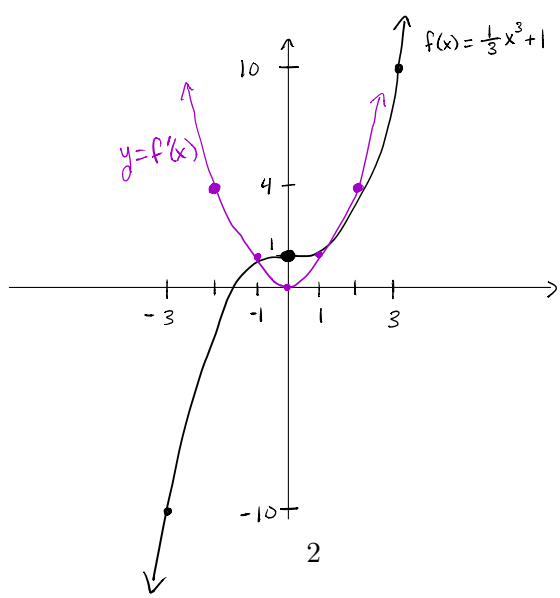
$$= \lim_{h \to 0} \frac{\cancel{x^2} + 2xh + h^2 - \cancel{x} - h - \cancel{2} - \cancel{x^2} + \cancel{x} + \cancel{2}}{h}$$

$$= \lim_{h \to 0} \frac{2x\cancel{h} + h^{\cancel{2}} - \cancel{h}^{1}}{\cancel{h}} = \lim_{h \to 0} 2x + h - 1 = 2x - 1$$

   (b) Do your results from this problem match your results from the last problem?

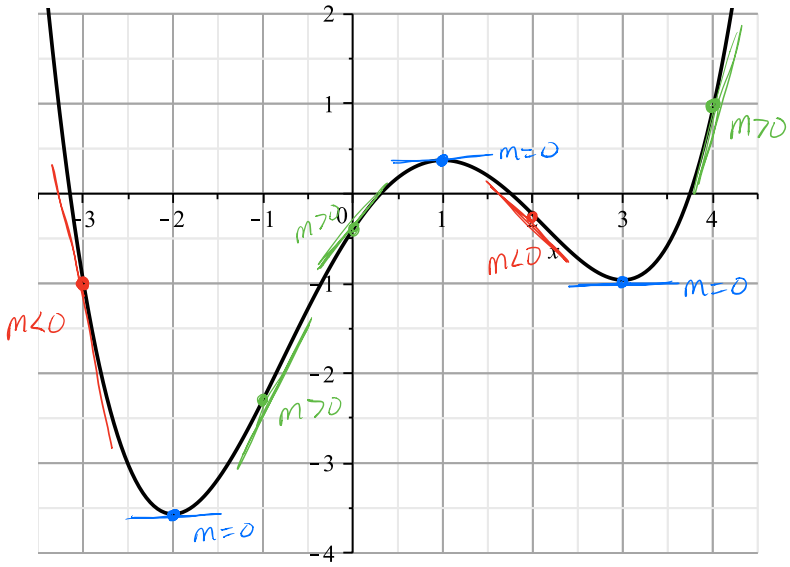
yes!

3. Use the same tool from Problem 1 to graph the function f(x)=1/3x^3+1 and its derivative, f'(x).



x	-2	-1	0	1	2
f'(x)	4	1	0	1	4

4. Below is a graph of a function.

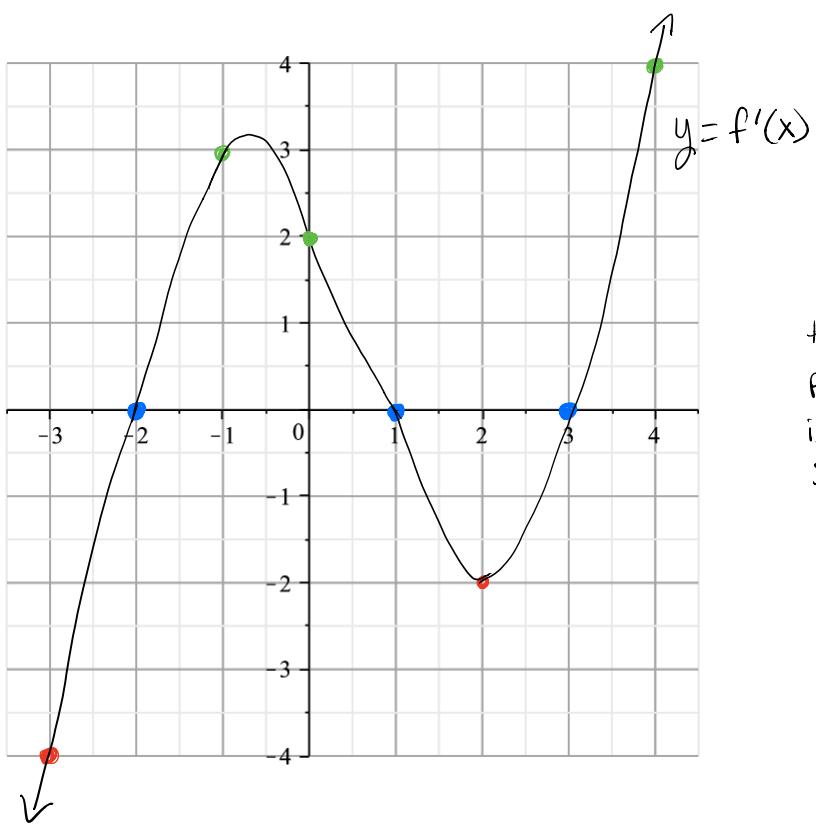


$f'(-2) = f'(1) = f'(3) = 0$

$f'(-3) < f'(2) < 0$

$f'(4) > f'(-1) > f'(0) > 0$

Without the aid of technology, use the graph of the function above to sketch a graph of its derivative function on the axes below.



Note that $y = f(x)$ is the graph of a degree 4 polynomial while $y = f'(x)$ is the graph of a degree 3 (or cubic) polynomial.

5. Based on your experience above, what seems to be true about the relationship between $f(x)$ and $f'(x)$?

   (a) Where $f(x)$ is increasing, $f'(x)$ is positive.

   (b) Where $f(x)$ is decreasing, $f'(x)$ is negative.

6. Below are some more involved questions. We will be addressing these in the coming sections. Do you have any guesses for these?

   (a) Where $f(x)$ is concave up, $f'(x)$ is increasing.

   (b) Where $f(x)$ is concave down, $f'(x)$ is decreasing.

   (c) Where $f(x)$ is discontinuous, undefined, or has a vertical tangent, $f'(x)$ is undefined.

   (d) What will the derivative be when $f(x)$ has a relative high point (maximum) or relative low point (minimum)?

   zero

   (e) If the derivative is 0 at a point, what are all the ways the original function could look?

   Maximum, minimum, constant

   (f) What about if the derivative has a relative maximum or minimum?

   The function switches concavity at these points.